21. LIGAMENT OF LILIEQUIST

DURATION : 07:37

STUDY SHEET

COURSE SUMMARY

This video explores the anatomy and mechanisms of the Liliequist ligament, an essential structure that acts as a diaphragm at the level of the third ventricle. It explains how this ligament interacts with the oculomotor nerve and the basilar trunk, thus influencing the ventricular system and the pituitary gland. The concepts of membranous tensions, cranial movement axes, and their impact on the facial and hormonal systems are discussed, while highlighting the importance of cranio-sternal-sacral and mandibular movements. The analysis of tensions can also help diagnose occlusal dysfunctions, offering therapeutic keys to restore bodily harmony.

THE LIGAMENT OF LILIEQUIST: ANATOMY AND MECHANISMS

The **ligament of Liliequist** is an anatomical structure that acts as a small diaphragm, commonly known as the **membrane of Liliequist**. This ligament inserts at the level of the **third ventricle** and descends to the **sphenoid**. Here, it plays a crucial role in relation to the **oculomotor nerve** and arterial passage, particularly the **basilar trunk**, which is responsible for the formation of the **cerebral arteries**.

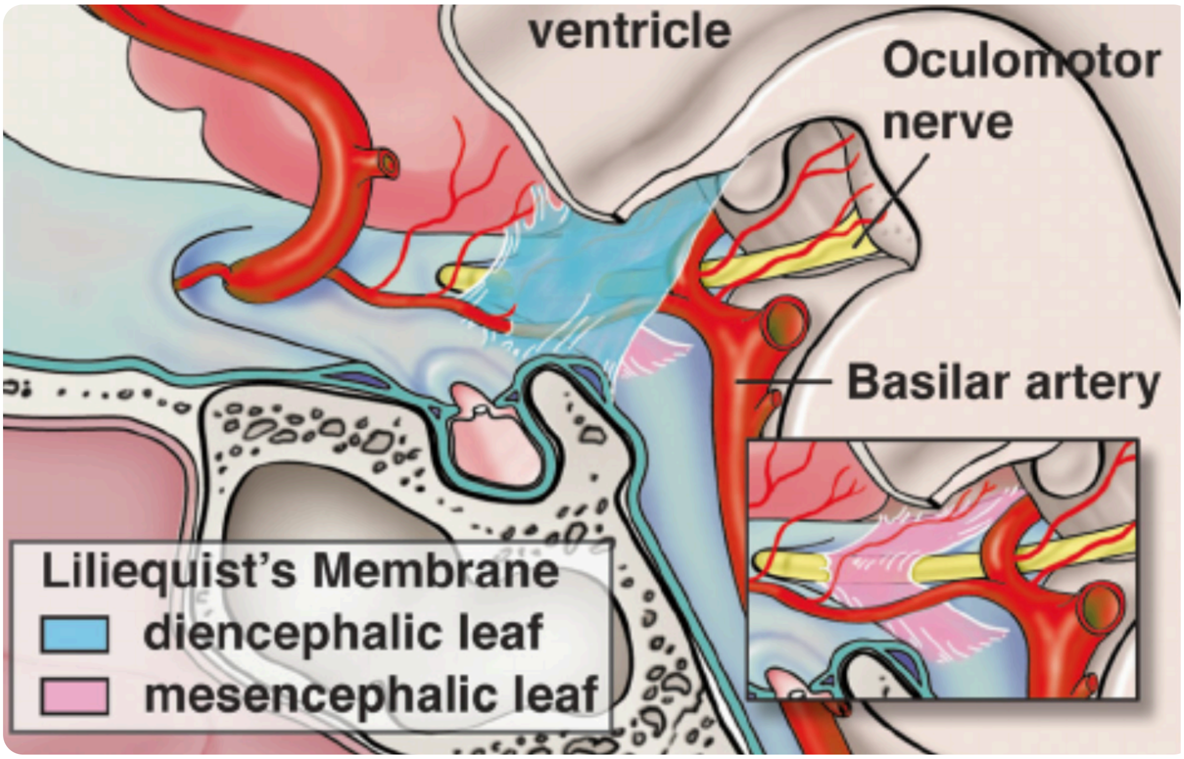

FIG. — Liliequist's ligament

As the ligament descends, it creates an **axis of tension** that influences the **ventricular system**. The third ventricle, the fourth ventricle, as well as the **lateral ventricles** are all interconnected by this dynamic. A tension pattern is established, having a significant impact on the space where the **pituitary gland** is located.

Above the pituitary gland is the **optic chiasm**, which is oriented at approximately 30 degrees, with a postural movement of 5 degrees. This movement is essential for **drainage** between the third ventricle, the sphenoid, and the surrounding **microvessels**. The **pericytes**, present in the microcapillaries, interact with the nerves, creating **blood-brain spaces** that act as a deep fulcrum.

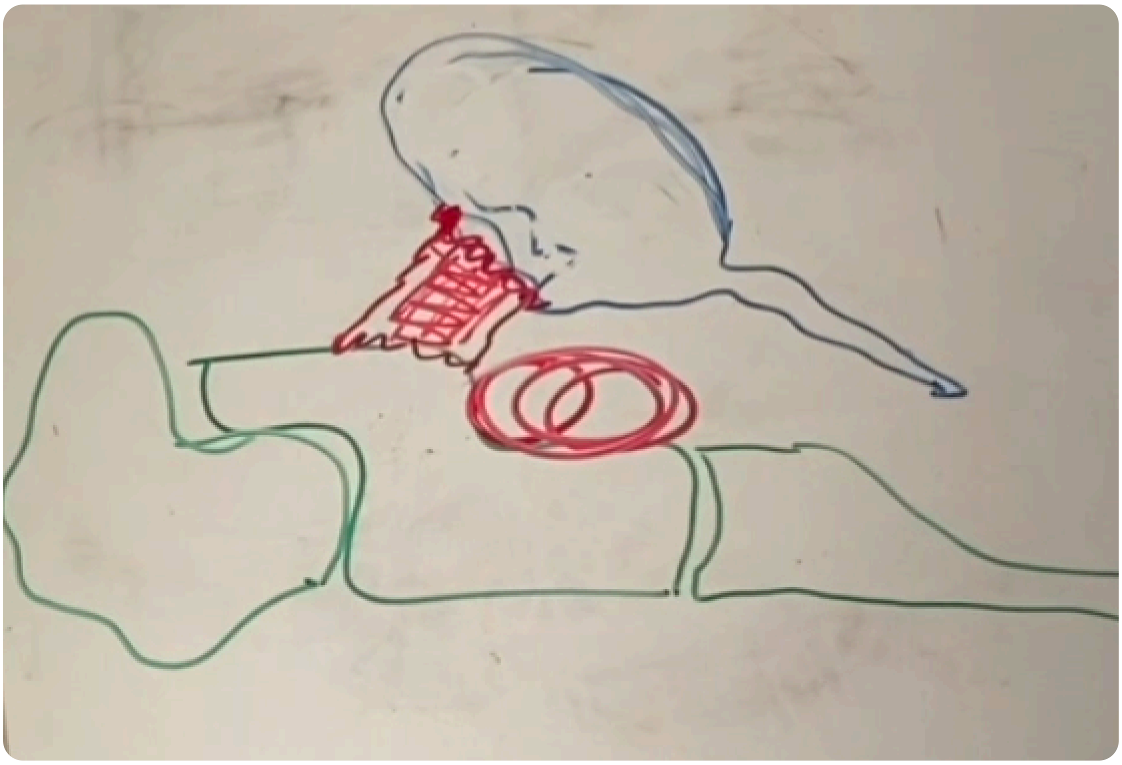

FIG. — Ventricular system

Examining the **presphenoids**, which correspond to the **occiput**, reveals a balance space on the membrane of Liliequist. This membranous tension affects both nerves and blood vessels, and is located near the pituitary gland, where the nervous system communicates with the hormonal system via the **hypothalamic-pituitary-thyroid-adrenal-gonadal axis**.

Cranial movement is also important to consider. By observing a skull, one can note the different movements at the level of the **parietals** and other structures. These movements follow specific lines of force. For example, the movement of **flexion** at the frontal level, associated with the **metopic suture**, represents two balance points that influence eye movement.

It is crucial to understand that the **brain** conditions the **ventricular system** and the **facial system**, both in terms of the **chondrocranium** (cartilage) and the **desmocranium** (membrane). During development, when the **occiput** descends, the **sacrum** also descends, and the **temporals** play a key role in this dynamic.

The **sternum** rises in response to these movements, converging towards the anterior midline. All of this is inscribed in the eye, and it is essential to add elements concerning the facial system and the capsular system of the eye, which insert around and organise the whole with a counter-rotation at the level of orbital development.

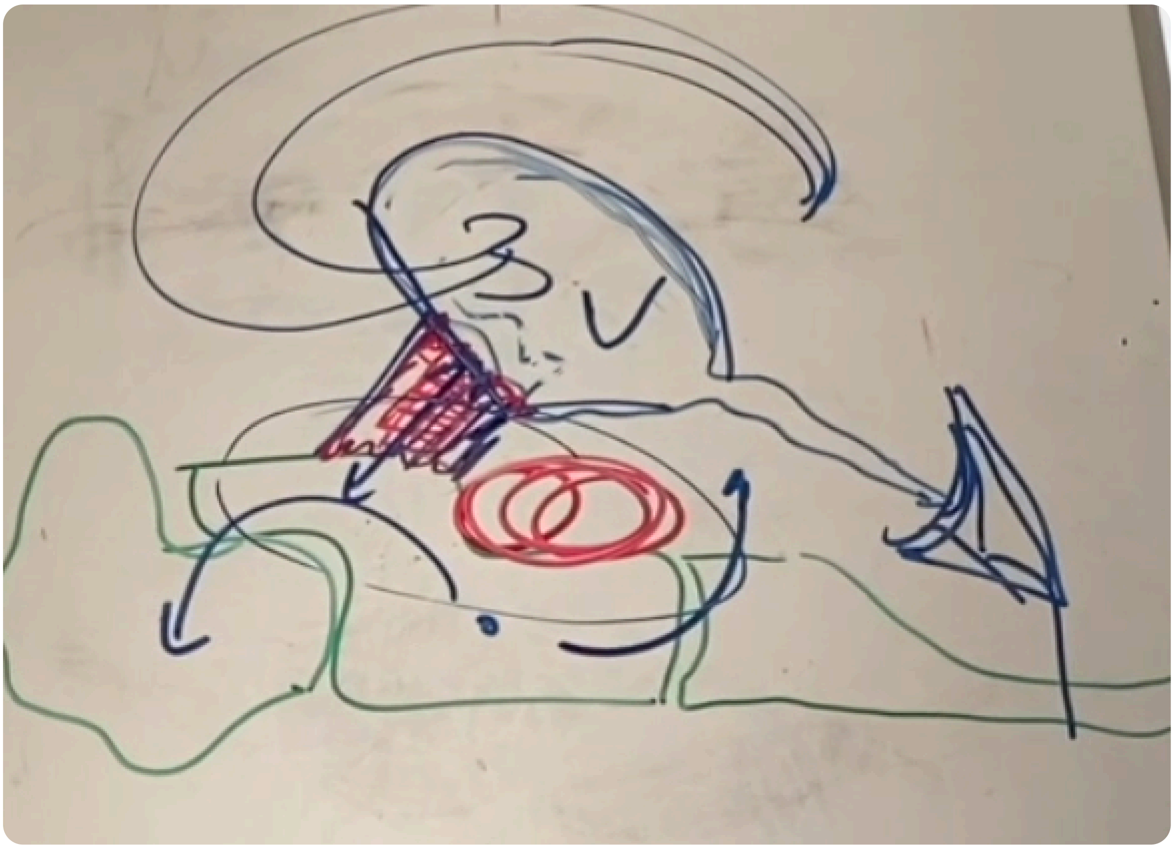

FIG. — Cranial movements

Craniosterno-sacral and mandibular movements are interconnected. The canine axes are also important, as they influence **occlusion**. If the upper and lower canine meeting is not correct, this leads to a loss of information at the brain level, because these electrical fields must unite.

Occlusion is therefore a reflection of the tensions present on the **sphenobasilar synchondrosis**. By observing the teeth, it is possible to determine the presence of **strain**, whether vertical, lateral or otherwise. This allows for an understanding of the constraints and the adjustment of necessary treatments to restore balance.

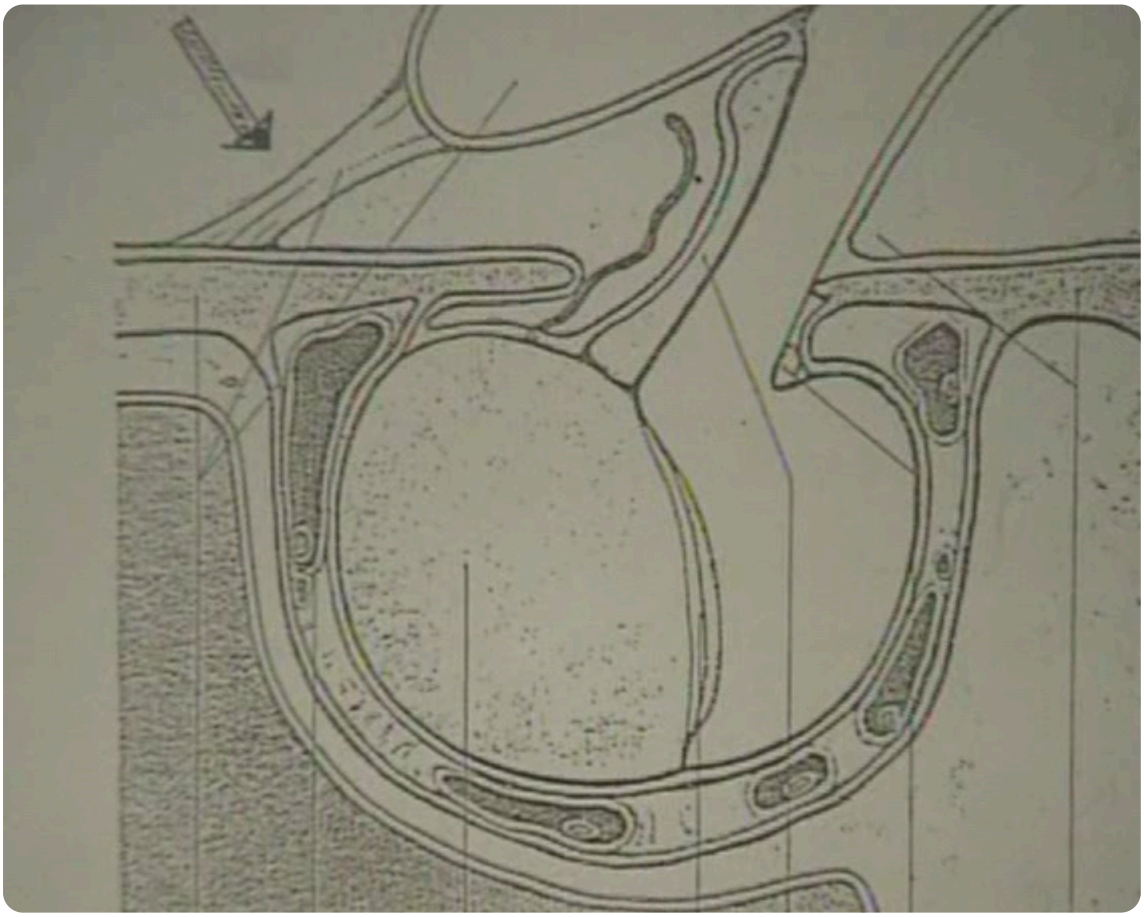

FIG. — Coordinated movements of the skull, sternum and sacrum